

# MALEK BENCHEHIDA

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## TECHNICAL SKILLS

- **Languages:** C++, Rust, C, Python, Go, Java, SQL, Assembly, JavaScript
- **Platforms & Hardware:** NVIDIA Jetson AGX Thor, Microcontrollers (ESP32/STM32), Linux Kernel, RTOS, ARM64, Docker, Git
- **Protocols & Interfaces:** CAN Bus, I2C, SPI, UART, LightGBM, REST APIs, Git Actions

## WORK EXPERIENCE

**MatchNice** Chicago, IL (Remote)  
*Software Engineer* May 2024 – Present

- Build a donation platform using Next.js 15, React 19, and TypeScript in a pnpm monorepo.
- Configure webhook interfaces and automate API type generation for high-performance data querying.
- Optimize Replit-based development pipelines to accelerate software delivery and testing cycles.

**CAS Enterprise Inc.** Chicago, IL  
*DevOps & Full Stack Developer* Aug 2023 – May 2024

- Managed AWS/Azure cloud infrastructure to ensure high availability for production applications.
- Automated CI/CD pipelines using Python to reduce deployment latency and increase shipping speed.
- Monitored system performance and hardware utilization to eliminate bottlenecks and backend downtime.

**University of Illinois Chicago (UIC)** Chicago, IL  
*Undergraduate Software Research Intern* Jun 2024 – Jul 2024

- Evaluated security vulnerabilities and execution risks in large language model implementations.
- Wrote Python validation scripts to test edge cases, monitor responses, and evaluate model robustness.

## PROJECT EXPERIENCE

**EdgeGuardNano Intrusion Detection System** 2026  
*Embedded Machine Learning Developer*

- Engineered a real-time, offline intrusion detection system deployed directly onto an NVIDIA Jetson AGX Thor edge device.
- Optimized LightGBM machine learning models for ARM64 CPU architecture, achieving a 632x throughput advantage over traditional cloud-based equivalents.
- Designed asynchronous data pipelines for high-throughput sensor monitoring and deterministic, low-latency execution.
- Developed an interactive dashboard to visualize real-time threat telemetry, model inference metrics, and edge system performance.

**SAE Formula Electric Team — UIC** 2024 – Present  
*Embedded Systems Software Engineer*

- Programmed real-time embedded control loops in C++ for motor speed and battery optimization.
- Migrated performance-critical firmware from C++ to Rust for memory safety and race-free concurrency.
- Developed communication interfaces between ECUs and sensors over the CAN Bus protocol.

**Reaction-Time Arena** 2026  
*Hardware & Embedded Systems Developer*

- Built a multiplayer hardware system utilizing three microcontrollers communicating via the I2C protocol.
- Designed a master-slave architecture to process low-latency inputs and execute real-time scoring logic.

**Real-Time Hand Gesture AI** 2025  
*Embedded & Computer Vision Developer*

- Trained a CNN model for Rock/Paper/Scissors hand gesture classification using TensorFlow and OpenCV.
- Implemented a low-latency game engine with camera inputs and real-time execution on local hardware.

## EDUCATION

**University of Illinois Chicago** Chicago, IL  
*Bachelor of Science in Computer Science* Expected Graduation: December 2026

- **Activities:** Technical Software Lead, SAE Formula Electric Team